COURSE TITLE	**BUSINESS INTELLIGENCE AND DATA STORYTELLING**
COURSE CODE	**04BM0101**
COURSE CREDITS	**3**

Course Outcomes: After completion of this course, student will be able to:

1. Demonstrate an understanding of descriptive analytics principles, Business Intelligence (BI) concepts, architecture, and tools.
2. Collect, clean, and prepare data for analysis using appropriate preprocessing techniques
3. Apply statistical methods and data mining to perform exploratory and descriptive data analysis.
4. Utilize modern analytics tools such as Python for business data analysis
5. Design and evaluate BI solutions, derive actionable insights from real-world and present findings to support decision-making

Pre-requisite of course: Basic knowledge of MS Excel and data handling

Teaching and Examination Scheme

Theory Hours	Tutorial Hours	Practical Hours	ESE	IA	CSE	Viva	Term Work
3	0	0	0	30	20	50	0

Contents : Unit	Topics	Contact Hours
1	**Descriptive Analytics Fundamentals** Definition, scope, key concepts (data, information, KPIs), evolution from traditional reporting to modern tools like Excel and R, Brief comparison with diagnostic, predictive, and prescriptive analytics, Business intelligence (BI) Overview – Components, Importance in decision-making, Descriptive Analytics in BI – Role within BI frameworks and workflows to support data interpretation and business insights, Data Models (Star Schema, Snowflake Schema), Decision Support Systems (DSS), Trends and Innovations in DSS, Marketing: Campaign ROI, market basket analysis	6
2	**Data Sources** Understand structured, unstructured, Data Collection & Preparation- Importance of data quality, cleansing, and preprocessing, Data transformation techniques, EDA Overview – Introduces the purpose and steps of exploratory data analysis for uncovering patterns and anomalies, Descriptive Statistics –Analyze data using measures of central tendency, Frequency distribution, percentiles, quartiles, dispersion, skewness, and kurtosis, Data Visualization using different plots.¬¬¬¬¬¬ Business Application: Healthcare- Patient analysis, Resource allocation and Utilization.	8

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Contents : Unit	Topics	Contact Hours
3	**Statistical Methods and Advanced Data Analysis** Measures of Central Tendency, Measures of Dispersion, visualizing descriptive Statistics, Histograms and box plots, Probability Distributions – Discrete Distributions– Binomial and Poisson distributions, Continuous Distributions – Normal and Exponential distributions, Fitting Data to Distributions, Hypothesis Testing Procedure, Statistical Tests – Z-test, t-test (One-Sample, Two-Sample), Chi-square test, and ANOVA, P-Value and Decision Rules, Business Application- Finance: Risk Management, fraud detection, portfolio analysis	15
4	**Data Visualization and Business Intelligence Tools** Principles of Data Visualization- importance, applications, Data Storytelling- Key concepts: narrative structure, core message, emotional engagement, Business Intelligence Tools- Power BI: Data Preparation & Modeling, Power BI Visualizations, PivotTables, PivotCharts, Reporting & Dashboard Development, Business case Studies: Sales Dashboard Implementation, HR analytics, Challenges, Solution	8
5	**Python in Data Analysis** Setting up Environment (Anaconda, Jupyter), Data Manipulation with pandas, Visualization with matplotlib, Advanced Visualization with seaborn, pandas + Visualization Integration, Real-World Projects: Hands-on Data Analysis, Best Practices: Coding & Data Management, Descriptive Analytics Applications- Retail: Sales trends, Inventory Management etc.	8
Total Hours		**45**

Textbook :

1 Business Analytics: Data Analysis & Decision Making, Albright, Winston, Cengage, 2020

2 Applied Business Analytics, Nathaniel Lin, FT Press, 2014

3 Practical Statistics for Data Scientist: 50+ Concepts Using R & Python, P. Bruce, A. Bruce, P. Gedeck, O'Reilly, 2020

4 Analyzing Data with Power BI and Power Pivot for Excelower Pivot for Excel, A. Ferrari, M. Russo, Microsoft Press, 2017

References:

1 Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking, Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking, F. Provost and T. Fawcett, O'Reilly Media, 2014

Suggested Theory Distribution:

The suggested theory distribution as per Bloom's taxonomy is as follows. This distribution serves as guidelines for teachers and students to achieve effective teaching-learning process

Distribution of Theory for course delivery and evaluation

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Remember / Knowledge	Understand	Apply	Analyze	Evaluate	Higher order Thinking / Creative
0.00	10.00	25.00	25.00	20.00	20.00

Instructional Method:

1 Use a project-based approach combining tool-driven labs (e.g., Power BI, Tableau), storytelling frameworks, real business dashboards, and peer presentations.

Supplementary Resources:

1 https://www.microsoft.com/en-us/power-platform/products/power-bi/downloads

2 https://www.boldbi.com/data-visualization-tool/lp/?utm_source=google_ads&utm_medium=&utm_campaign=boldbi_search_india_buyer_intent_gadwJul2025&adgroup=166208920485&adpos=&utm_term=data%20visualization&creative=749349514037&extension=&gad_source=1&gad_campaignid=21733269773&gbraid=0AAAAA-NRc_BdqhHWDJCEKMbJpNkJnHEm7&gclid=CjwKCAjw4K3DBhBqEiwAYtG_9EKAgTwPi61-So0V9WUtgeTIsf3fm4GKyWxHeJxFoMZVShc2YKjREROcotoQAvD_BwE

3 https://www.tableau.com/visualization/what-is-data-visualization

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